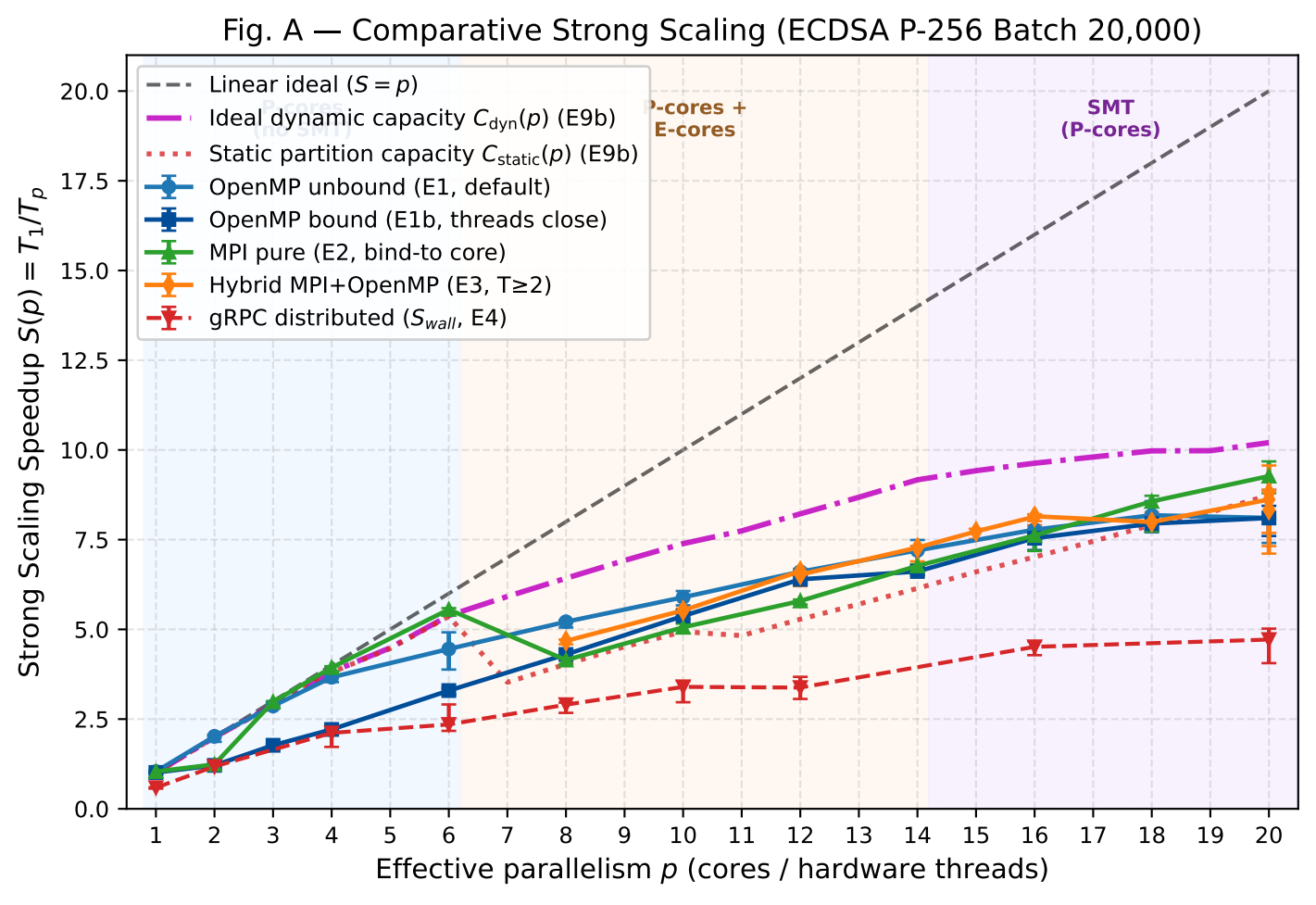
Fig. A — Comparative Strong Scaling (ECDSA P-256 Batch 20,000)
Linear ideal ($S = p$)
Ideal dynamic capacity $C_{dyn}(p)$ (E9b)
Static partition capacity $C_{static}(p)$ (E9b)
OpenMP unbound (E1, default)
OpenMP bound (E1b, threads close)
MPI pure (E2, bind-to core)
Hybrid MPI+OpenMP (E3, T≥2)
gRPC distributed (S_{wall}, E4)
P-cores +
E-cores
SMT
(P-cores)
Strong Scaling Speedup $S(p) = T_1/T_p$
Effective parallelism p (cores / hardware threads)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20